

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Enterprise Java

M1:Practical –VI

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Class: TYIT	Batch: 1
Date of Assignment: 22/08/2026	Date/Time of Submission: 22/08/2026

6. Implement the following JSP JSTL and EL Applications:

a. Create an html page with fields, eno, name, age, designation, salary. Now on submit this data to a JSP page which will update the employee table of database with matching eno.

index.jsp:

```
<%@ page language="java" contentType="text/html; charset=UTF-8" %>
<%@ page import="java.util.Date" %>
<html>
  <head>
    <title>JSP Directives and Implicit Objects</title>
  </head>
  <body>
    <h2>JSP Directives and Implicit Objects</h2>
    <h3>JSP Directives</h3>
    <p>Language: Java</p>
    <p>Content Type: <%= response.getContentType()%></p>
    <p>Current Date: <%= new Date()%></p>
    <h3>Implicit Objects</h3>
    <p>1. Request Method: <%= request.getMethod()%></p>
    <p>2. Response Content Type: <%= response.getContentType()%></p>
    <p>3. Session ID: <%= session.getId()%></p>
    <p>4. Application Server: <%= application.getServerInfo()%></p>
    <p>5. Page Name: <%= page.getClass().getName()%></p>
    <p>6. PageContext: <%= pageContext.getClass().getName()%></p>
    <p>7. Request Parameter: <%= request.getParameter("name")%></p>
    <p>8. Config Servlet Name: <%= config.getServletName()%></p>
    <p>9. Exception Object: Exception object is available only in error pages.</p>
    <h2>Asmitha T002</h2>
  </body>
</html>
```

Output:

JSP Directives and Implicit Objects

JSP Directives

Language: Java

Content Type: text/html; charset=UTF-8

Current Date: Sat Aug 22 11:28:41 IST 2026

Implicit Objects

1. Request Method: GET
2. Response Content Type: text/html; charset=UTF-8
3. Session ID: 80b6f5ddbfa94e358e4a51c7f8e0
4. Application Server: GlassFish Server Open Source Edition 4.1.1
5. Page Name: org.apache.jsp.index_jsp
6. PageContext: org.apache.jasper.runtime.PageContextImpl
7. Request Parameter: null
8. Config Servlet Name: jsp
9. Exception Object: Exception object is available only in error pages.

Asmitha T002

b. Create a JSP page to demonstrate the use of Expression language.**index.jsp:**

```

<html>
<head>
<title>Gym Membership Form</title>
</head>
<body>
<h2>Gym Membership Form</h2>
<form action="gymprocess.jsp" method="post">
    Name:
    <input type="text" name="name">
    <br><br>
    Age:
    <input type="text" name="age">
    <br><br>
    Email:
    <input type="text" name="email">
    <br><br>
    Mobile:
    <input type="text" name="mobile">
    <br><br>
    Gender:
    <input type="radio" name="gender"
value="Male"> Male
    <input type="radio" name="gender"
value="Female"> Female
    <br><br>
    Membership Type:
    <select name="membership">
        <option value="">Select
Membership</option>
        <option value="Monthly">Monthly</option>
        <option
value="Quarterly">Quarterly</option>
        <option value="Yearly">Yearly</option>
    </select>
    <br><br>
    Fitness Goal:
    <input type="checkbox" name="goal"
value="Weight Loss"> Weight Loss
    <input type="checkbox" name="goal"
value="Muscle Gain"> Muscle Gain
    <input type="checkbox" name="goal"
value="Fitness"> Fitness
    <br><br>
    Address:
    <textarea name="address"></textarea>
    <br><br>
    <input type="submit" value="Register">
    </form>
</body>
</html>

```

gymprocess.jsp:

```

<html>
<head>
<title>Gym Membership Details</title>
</head>
<body>
<h2>Gym Membership Details</h2>
<%
    String name = request.getParameter("name");
    String age = request.getParameter("age");
    String email = request.getParameter("email");
    String mobile =
request.getParameter("mobile");

```

```

    String gender =
request.getParameter("gender");
    String membership =
request.getParameter("membership");
    String address =
request.getParameter("address");
    String[] goals =
request.getParameterValues("goal");
    boolean valid = true;
    if (name == null || name.trim().equals("")) {
        out.println("Name is required.<br>");
        valid = false;
    }
    if (age == null || age.trim().equals("")) {
        out.println("Age is required.<br>");
        valid = false;
    }
    if (email == null || email.trim().equals("")) {
        out.println("Email is required.<br>");
        valid = false;
    }
    if (mobile == null || mobile.trim().equals("")) {
        out.println("Mobile number is
required.<br>");
        valid = false;
    }
    if (gender == null) {
        out.println("Please select gender.<br>");
        valid = false;
    }
    if (membership == null ||
membership.equals("")) {
        out.println("Please select membership
type.<br>");
        valid = false;
    }
    if (goals == null) {
        out.println("Please select at least one
fitness goal.<br>");
        valid = false;
    }
    if (address == null || address.trim().equals(""))
    {
        out.println("Address is required.<br>");
        valid = false;
    }
    if (valid) {
        %>
        <h3>Gym Membership Registration
Successful</h3>
        Name: <%= name%><br><br>
        Age: <%= age%><br><br>
        Email: <%= email%><br><br>
        Mobile: <%= mobile%><br><br>
        Gender: <%= gender%><br><br>
        Membership Type: <%=
membership%><br><br>
        Fitness Goals:
        <%
            for (String goal : goals) {
                out.println(goal + " ");
            }
        %>
        <br><br>

```

```

Address: <%= address%><br><br>
<%
}

```

```

%>
</body>
</html>

```

Output:**Gym Membership Form**

Name:

Age:

Email:

Mobile:

Gender: ☐ Male ☒ Female

Membership Type:

Fitness Goal: ☐ Weight Loss ☐ Muscle Gain ☒ Fitness

Address:

Gym Membership Details**Gym Membership Registration Successful**

Name: Asmitha

Age: 21

Email: asmi.com

Mobile: 1234567892

Gender: Female

Membership Type: Yearly

Fitness Goals: Fitness

Address: Aarey colony

c. Create a JSP application to demonstrate the use of JSTL.**sql code:**

```

CREATE DATABASE ecommerce;
USE ecommerce;
Database changed
CREATE TABLE product (
    product_id INT PRIMARY KEY,
    product_name VARCHAR(100),
    category VARCHAR(50),
    brand VARCHAR(50),
    price DECIMAL(10,2),
    quantity INT,
    rating DECIMAL(2,1),
    discount DECIMAL(5,2),
    description VARCHAR(255)
);

```

index.jsp:

```

<%@ page language="java"
    contentType="text/html; charset=UTF-8"
    import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>E-Commerce Products</title>
<style>
    body {

```

```

font-family: Arial, sans-serif;
background-color: #f2f2f2;
margin: 30px;
}
h2 {
    text-align: center;
    color: #333;
}
table {
    width: 100%;
    border-collapse: collapse;
    background-color: white;
}
th {
    background-color: #2874A6;
    color: white;
    padding: 12px;
}
td {
    padding: 10px;
    border: 1px solid #ddd;
    text-align: center;
}
tr:nth-child(even) {
    background-color: #f2f2f2;
}
tr:hover {
    background-color: #D6EAF8;
}
</style>
</head>
<body>
<h2>E-Commerce Product List</h2>
<table>

```

```

<tr>
  <th>Product ID</th>
  <th>Product Name</th>
  <th>Category</th>
  <th>Brand</th>
  <th>Price</th>
  <th>Quantity</th>
  <th>Rating</th>
  <th>Discount</th>
  <th>Description</th>
</tr>
<%
  Connection con = null;
  Statement stmt = null;
  ResultSet rs = null;
  try {
    Class.forName("com.mysql.jdbc.Driver");

    con = DriverManager.getConnection(
"jdbc:mysql://localhost:3306/ecommerce",
    "root",
    "root"
);
    stmt = con.createStatement();

    rs = stmt.executeQuery("SELECT *
FROM product");
    while (rs.next()) {
      <tr>
        <td><%= rs.getInt("product_id")%></td>
        <td><%=
rs.getString("product_name")%></td>
        <td><%= rs.getString("category")%></td>
        <td><%= rs.getString("brand")%></td>
        <td><%= rs.getDouble("price")%></td>
        <td><%= rs.getInt("quantity")%></td>
        <td><%= rs.getDouble("rating")%> ★ </td>
        <td><%=
rs.getDouble("discount")%>%</td>
        <td><%= rs.getString("description")%></td>
      </tr>
    }
  } catch (Exception e) {
    out.println("<p>Error: " + e.getMessage()
+ "</p>");
  } finally {
    try {
      if (rs != null) {
        rs.close();
      }
      if (stmt != null) {
        stmt.close();
      }
      if (con != null) {
        con.close();
      }
    } catch (Exception e) {
      e.printStackTrace();
    }
  }
}
%>
</table>
</body>
</html>

```

```

mysql> SELECT * FROM product;
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| product_id | product_name | category | brand | price | quantity | rating | discount | description |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
| 1 | iPhone 15 | Mobile | Apple | 69999.00 | 10 | 4.8 | 10.00 | Apple iPhone with advanced camera and performance |
| 2 | Galaxy S24 | Mobile | Samsung | 64999.00 | 15 | 4.7 | 12.00 | Samsung smartphone with AMOLED display |
| 3 | MacBook Air M2 | Laptop | Apple | 89999.00 | 8 | 4.9 | 8.00 | Lightweight laptop powered by Apple M2 chip |
| 4 | WH-1000XM5 | Headphones | Sony | 29999.00 | 20 | 4.6 | 15.00 | Wireless noise cancelling headphones |
| 5 | Inspiron 15 | Laptop | Dell | 55999.00 | 12 | 4.5 | 10.00 | Dell laptop suitable for everyday computing |
+-----+-----+-----+-----+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

```

E-Commerce Product List

Product ID	Product Name	Category	Brand	Price	Quantity	Rating	Discount	Description
1	iPhone 15	Mobile	Apple	₹69999.0	10	4.8 ★	10.0%	Apple iPhone with advanced camera and performance
2	Galaxy S24	Mobile	Samsung	₹64999.0	15	4.7 ★	12.0%	Samsung smartphone with AMOLED display
3	MacBook Air M2	Laptop	Apple	₹89999.0	8	4.9 ★	8.0%	Lightweight laptop powered by Apple M2 chip
4	WH-1000XM5	Headphones	Sony	₹29999.0	20	4.6 ★	15.0%	Wireless noise cancelling headphones
5	Inspiron 15	Laptop	Dell	₹55999.0	12	4.5 ★	10.0%	Dell laptop suitable for everyday computing